claim; $J_1'(0) = 1/2$, $J_n'(0) = 0$

proof $\exp\left[\frac{1}{2}(x)(t-1/t)\right] = \sum_{-\infty}^{\infty} J_n(x)\, t^n$

take $\partial/\partial x$

$$\frac{1}{2}(t-1/t)\exp\left\{\frac{1}{2}(x)(t-1/t)\right\} = \sum_{-\infty}^{\infty} J_n'(x)\, t^n$$

$x \longrightarrow 0$

$$\Rightarrow \quad \frac{1}{2}(t-\frac{1}{t}) = \sum_{-\infty}^{\infty} J_n'(0)\, t^n$$

$J_1'(0) = \frac{1}{2}$, $J_{-1}'(0) = -\frac{1}{2}$

$J_n(0) = 0$.

claim: $\frac{d}{dx} J_p(kx) = -k J_{p+1}(kx) + \frac{p}{x} J_p(kx) \quad k>0$

proof:

$$J_p(kx) = \sum_{m=0}^{\infty} \frac{(-1)^m (kx/2)^{2m+p}}{m!\,\Gamma(m+p+1)}$$

$$\frac{d}{dx} J_p(kx) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \frac{(k/2)^{2m+p}}{\Gamma(m+p+1)} (2m+p) x^{2m+p-1}$$

$$= 2 \sum_{m=0}^{\infty} \frac{(-1)^m}{(m-1)!} \left(\frac{k}{2}\right)^{2m+p} \frac{x^{2m+p-1}}{\Gamma(m+p+1)}$$

$$+ \frac{p}{x} \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \frac{1}{\Gamma(m+p+1)} \left(\frac{kx}{2}\right)^{2m+p}$$

$$= 2 \sum_{m=0}^{\infty} \frac{(-1)^m}{(m-1)!} \left(\frac{k}{2}\right)^{2m+p} \frac{x^{2m+p-1}}{\Gamma(m+p+1)}$$

$$+ \frac{p}{x} J_p(kx)$$

$$= 2\left(\frac{k}{2}\right) \sum_{m=0}^{\infty} \frac{(-1)^m m}{(m)!} \left(\frac{kx}{2}\right)^{2m+p-1} \times \frac{1}{\Gamma(m+p+1)}$$

$$= k \sum_{m=1}^{\infty} \frac{(-1)^m m}{(m)!} \left(\frac{kx}{2}\right)^{2m+p-1} \frac{1}{\Gamma(m+p+1)}$$

$$m \longrightarrow m+1$$

$$= k \sum_{m=0}^{\infty} \frac{(-1)^{m+1}}{m!} \left(\frac{kx}{2}\right)^{2m+p+1} \frac{1}{\Gamma(m+p+2)}$$

$$= -k \; J_{p+1}$$

claim: $$J_p = \frac{(x/2)^p}{\sqrt{\pi}\,\Gamma(p+1/2)} \int_{-1}^{1} (1-t^2)^{p-1/2} e^{ixt}\,dt$$

proof: $$\int_{-1}^{1} (1-t^2)^{p-1/2} e^{ixt}\,dt$$

$$= 2\int_{0}^{1} (1-t^2)^{p-1/2} \underbrace{\cos xt}_{\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n} t^{2n}}{(2n)!}}\,dt$$

$$= 2\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} \underbrace{\int_0^1 (1-t^2)^{p-1/2} t^{2n}\,dt}_{\text{Beta fun}}$$

$$\int_0^1 (1-t^2)^{p-1/2} t^{2n}\,dt$$

$$u = t^2$$

$$du = 2t\,dt \quad \Rightarrow \quad dt = \frac{1}{2} u^{-1/2}\,du$$

$$\Rightarrow \frac{1}{2}\int_0^1 (1-u)^{p-1/2} u^{n-1/2}\,du$$

$$= \frac{1}{2} B(p+1/2, n+1/2)$$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} B(p+1/2, n+1/2)$$

$$B(p+1/2, n+1/2) = \frac{\Gamma(p+1/2)\,\Gamma(n+1/2)}{\Gamma(n+p+1)}$$

Since $\Gamma(n+1/2) = \frac{\sqrt{\pi}\,(2n)!}{2^{2n}\,(n)!}$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{(-1)^n \sqrt{\pi}\,(2n)!}{(2n)!\;(n)!\;\Gamma(n+p+1)} \Gamma(p+1/2)\, \frac{x^{2n}}{2^{2n}}\, \frac{(x/2)^p}{(x/2)^p}$$

$$= \frac{\Gamma(p+1/2)\sqrt{\pi}}{(x/2)^p} \underbrace{\sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(n+p+1)} \left(\frac{x}{2}\right)^{2n+p}}_{J_p(x)}$$

Integrals main points

$$\frac{d}{dx}\left[x^p J_p(x)\right] = x^p J_{p-1}$$

$$\frac{d}{dx}\left[x^{-p} J_p\right] = -x^{-p} J_{p+1}$$

$$\Rightarrow \int x^p J_{p-1} = x^p J_p + C$$

$$\int x^{-p} J_{p+1} = -x^{-p} J_p + C$$

$$\int x^m J_n \qquad x^m \underline{x^{n+1}} x^{-n-1} \underline{J_n}$$

$$u = x^{m-n-1} \qquad v = x^{n+1} J_{n+1}$$

$$du = (m-n-1)\, x^{m-n-2} \qquad dv = x^{n+1} J_n$$

$$x^m J_{n+1} - (m-n-1) \int x^{m-1} J_{n+1}$$

$$I(m,n) \longrightarrow I(m-1, n+1)$$

$$x^m \underline{x^{1-n}} x^{n-1} \underline{J_n} \qquad p = n-1$$

$$u = x^{m+n-1} \qquad v = -x^{1-n} J_{n-1}$$

$$du = (m+n-1)\, x^{m+n-2} \qquad dv = x^{1-n} J_n$$

$$= -x^m J_{n-1} + (m+n-1) \int x^{m-1} J_{n-1}$$

$$I(m,n) \longrightarrow I(m-1, n-1)$$

$$x^m J_n = x^m x^n x^{-n} J_n$$

$$u = x^n J_n \qquad v = \frac{x^{m-n+1}}{(m-n+1)}$$

$$du = x^n J_{n-1} \qquad dv = x^{m-n}$$

$$= x^{m+1} \frac{J_n}{(m-n+1)} - \frac{1}{(m-n+1)} \int x^{m+1} J_{n-1}$$

$$I(m,n) \longrightarrow I(m+1, n-1)$$

$u = x^{-n} J_n$ $\qquad v = \frac{x^{m+n+1}}{(m+n+1)}$

$du = -x^{-n} J_{n+1}$ $\qquad dv = x^{m+n}$

$$\frac{x^{m+1} J_n}{(m+n+1)} + \frac{1}{(m+n+1)} \int x^{m+1} J_{n+1}$$

$$I(m,n) \longrightarrow I(m+1, n+1)$$

$I(m+1, n+1)$

$$I(m+1, n-1) \longleftarrow I(m,n) \longrightarrow I(m-1, n+1)$$

$$\downarrow$$

$$I(m-1, n-1)$$

evaluate : $\int x^2 J_0 = x^2 J_1 + x J_0 - \int J_0$

$(2,0) \longrightarrow (1,1) \longrightarrow (0,0)$

- $x^m J_{n+1} - (m-n-1) \int x^{m-1} J_{n+1}$
- $-x^m J_{n-1} + (m+n-1) \int x^{m-1} J_{n-1}$

$$x^2 J_1 - \left[\int x^1 J_1 \right]$$

$$x^2 J_1 - \left[-x J_0 + \int J_0 \right]$$

$$= x^2 J_1 + x J_0 - \int J_0 \quad \checkmark$$

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evaluate: $\int x^3 J_0 \, dx = (x^3 - 4x) J_1 + 2x^2 J_0$

$(3, 0) \longrightarrow (2, 1) \longrightarrow (1, 0)$

$$x^3 J_1 - (2) \left[\int x^2 J_1 \right]$$

$$x^3 J_1 - 2 \left[-x^2 J_0 + 2 \int x J_0 \right]$$

$$x^3 J_1 + 2x^2 J_0 - 4 x J_1 + C \quad \checkmark$$

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claim:

$$\int x^{-2} J_2 \, dx = -\frac{2}{3x^2} J_1 - \frac{1}{3} J_1 \to \frac{1}{3x} J_0 + \frac{1}{3} \int J_0$$

proof: $(-2, 2) \longrightarrow (-1, 1) \longrightarrow (0, 0)$

$$x^{m+1} \frac{J_n}{(m-n+1)} - \frac{1}{(m-n+1)} \int x^{m+1} J_{n-1}$$

$$= \frac{x^{-1}}{(-3)} J_2 + \frac{1}{3} \left[- J_1 + \int J_0 \right]$$

$$= -\frac{1}{3} \frac{J_2}{x} - \underline{\frac{1}{3} J_1} + \underline{\frac{1}{3} \int J_0}$$

$$\frac{J_p(2p)}{x} = J_{p-1} + J_{p+1} \qquad p = 1$$

$$\frac{2 J_1}{x} = J_0 + J_2 \Rightarrow J_2 = \frac{2 J_1}{x} - J_0$$

Definite Integrals involving Bessel function

claim: $\int_0^\infty e^{-st}\, t^{p/2}\, J_p(2\sqrt{t})\, dt$

$$= \frac{1}{s^{p+1}} e^{-1/s} \qquad p > -1/2\ ,\ s > 0$$

proof: $t^{p/2} J_p(2\sqrt{t}) = t^{p/2} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k+p+1)} (t^{1/2})^{2k+p}$

$$= \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k+p+1)} t^{k+p}$$

$$\Rightarrow \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k+p+1)} \int_0^\infty e^{-st}\, t^{k+p}\, dt$$

$u = st \qquad s > 0$

$du = s\, dt$

$$\int_0^\infty e^{-u}\, s^{-k-p-1}\, u^{k+p}\, du$$

$$= \frac{1}{s^{k+p+1}} \underbrace{\int_0^\infty e^{-u}\, u^{k+p}\, du}_{\Gamma(k+p+1)}$$

$$= \frac{\Gamma(k+p+1)}{s^{k+p+1}}$$

$$\Rightarrow \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\, s^{k+p+1}} = \frac{1}{s^{p+1}} e^{-1/s}$$

claim: $\int_0^\infty e^{-ax} x^p J_p(bx)\, dx = \frac{(2b)^p \Gamma(p+1/2)}{\sqrt{\pi}\,(a^2+b^2)^{p+1/2}}$

} proof in TexT & in Assignment

$p = 0$

$$\int_0^\infty e^{-ax} J_0(bx)\, dx = \frac{1}{\sqrt{a^2+b^2}}$$

claim: $\int_0^\infty e^{-ax} x^2 J_0(bx)\, dx = \frac{2a^2-b^2}{(a^2+b^2)^{5/2}}$

proof:

$$\int_0^\infty \frac{d^2}{da^2}\left(e^{-ax}\right) J_0(bx)\, dx$$

$$= \frac{d^2}{da^2} \int_0^\infty J_0(bx)\, e^{-ax}\, dx$$

$$= \frac{d^2}{da^2}\left[(a^2+b^2)^{-1/2} \right]$$

$$= \frac{d}{da}\left[-\frac{1}{2}(a^2+b^2)^{-3/2}(2a) \right]$$

$$= -\frac{d}{da}\left[a(a^2+b^2)^{-3/2} \right]$$

$$= -\left[\frac{1}{(a^2+b^2)^{3/2}} - 3\frac{a^2}{(a^2+b^2)^{5/2}} \right]$$

$$= -\left[\frac{(a^2+b^2) - 3a^2}{(a^2+b^2)^{5/2}} \right]$$

$$= \frac{2a^2 - b^2}{(a^2+b^2)^{5/2}}$$ ✓

claim: $$\int_0^\infty e^{-ax^2} x^{p+1} J_p(bx)\,dx = \frac{b^p}{(2a)^{p+1}} e^{-b^2/4a}$$

proof: $$J_p(bx) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(p+k+1)} \left(\frac{bx}{2}\right)^{2k+p}$$

$$\Rightarrow \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k+p+1)} \left(\frac{b}{2}\right)^{2k+p} \int_0^\infty e^{-ax^2} x^{2k+2p+1}\,dx$$

$$u = ax^2 \quad \Rightarrow \quad x = (u/a)^{1/2}$$

$$du = a\,2x\,dx \quad \Rightarrow \quad dx = \frac{du}{2ax} = \left(\frac{u}{a}\right)^{-1/2} \frac{du}{2a}$$

$$\Rightarrow \int_0^\infty e^{-ax^2} x^{2k+2p+1}\,dx$$

$$= \int_0^\infty e^{-u} \left(\frac{u}{a}\right)^{k+p+1/2} \left(\frac{u}{a}\right)^{-1/2} \frac{du}{2a}$$

$$= \frac{1}{2a\,a^{k+p}} \underbrace{\int_0^\infty e^{-u} u^{k+p}\,du}_{\Gamma(k+p+1)}$$

$$= \frac{\Gamma(k+p+1)}{2\,a^{k+p+1}}$$

$$\Rightarrow \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\,\Gamma(k+p+1)} \left(\frac{b}{2}\right)^{2k+p} \times \frac{\Gamma(k+p+1)}{2\,a^{k+p+1}}$$

$$= \sum_{k=0}^{\infty} \frac{(-1)^k \left(b^2/4a\right)^k}{k!} \times \frac{b^p}{2^{p+1}\,a^{p+1}}$$

$$= \frac{b^p}{2^{p+1}\,a^{p+1}}\, e^{-b^2/4a} \quad \checkmark$$

claim: $$\int_0^\infty \frac{\sin x}{x} J_0(bx)\,dx = \sin^{-1}\frac{1}{b}$$

proof: $$\int_0^\infty \cos(ax)\, J_0(bx)\,dx = \frac{1}{\sqrt{b^2-a^2}}$$

$$\int_0^a \cos(a'x)\,da' = \frac{\sin(ax)}{x}$$

$$\Rightarrow \int_0^\infty \frac{\sin(ax)}{x} J_0(bx)\,dx = \int_0^a \frac{da'}{\sqrt{b^2-a'^2}}$$

$$a' = b\sin(\theta)$$

$$da' = b\cos\theta\, d\theta$$

$$\int \frac{da'}{\sqrt{b^2-a'^2}} = \int d\theta = \theta = \sin^{-1}\left(\frac{a'}{b}\right)\Big|_0^a$$

$$= \sin^{-1}\left(\frac{a}{b}\right)$$

$$\Rightarrow \int_0^\infty \frac{\sin(ax)}{x} J_0(bx)\,dx = \sin^{-1}\left(\frac{a}{b}\right)$$